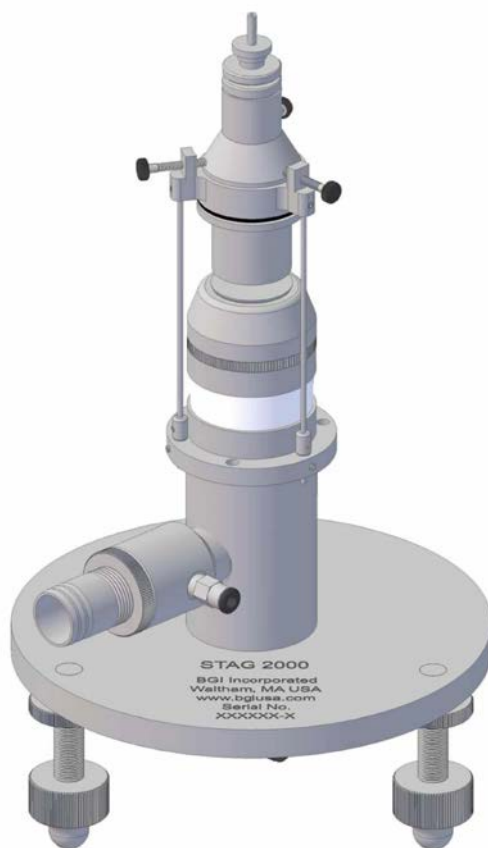


Instruction Manual

STAG 2000

Spinning Top Monodisperse Aerosol generator



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DISCLAIMER

The STAG 2000 is a monodisperse aerosol generator with a history dating back to the original work of Walton and Prewett in 1945⁽¹⁾. In its operation, it utilizes an unrestrained spinning top. It is an instrument for the expert aerosol scientist who needs to make large quantities of monodisperse aerosols. There are no hazards inherent in its use or application. Nevertheless, BGI Incorporated assumes no liability for damage or injury involved in its utilization.

It is also recognized that it may be applied to the aerosolization of hazardous, toxic or sinister materials. BGI Incorporated specifically disclaims any responsibility for any problems arising from the use of hazardous materials.

Instruments requiring factory repair must be sent to BGI Incorporated, under an RMA number, completely disassembled with a certification of decontamination.

Because of the attention to detail required in the operation of the instrument the BGI Inc. standard warranty is limited to defects in original workmanship. Inability to generate monodisperse aerosols, on the part of a user, is specifically not covered.

Warning

Read this message before attempting to use the instrument.

Two parts of this device are not covered by warranty.

1. Part No. A2214 Rotor Disc

We have found that technical personnel will drop the rotor causing a small dent on the sharp edge and degrading the ability of the device to produce a monodisperse aerosol. This situation frequently goes unnoticed. All Rotor Discs are given a careful final inspection prior to shipment.

2. Part No. A2222 Dampner

This part has an effectively unlimited lifetime if it does not get torn or ripped. The only way that this can occur is if the instrument is lifted by the spider assembly on top

(ALWAYS LIFT INSTRUMENT FROM BASE PLATE)

or disassembly of the instrument is attempted without due attention to the instructions. The STAG 2000 is carefully packed and the dampner will not rip during shipment. The lack of due care during the unpacking process will cause damage. Because the instrument can not function without this critical component, each new unit is furnished with a spare dampner.

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1.0 Introduction

The BGI STAG 2000 is a new design, air driven, monodisperse aerosol generator which, continues the decades long development work of our predecessors. It is also the first time an enclosure is commercially available.

There are numerous references on the design use and application of devices in this class and a web search will provide many. Of particular utility is the search engine: <http://www.google.com> . While this instruction manual is more extensive than any heretofore provided with an instrument of this type it does not contain every last detail for the successful operation of the unit. The seminal references that should be consulted, for the successful operation of this generator are:

1. Walton, W.H. and W.C. Prewett, "The Production of Sprays and Mists of Uniform Drop Size by Means of Spinning Disc Type Sprayers". The Proc. Phys. Soc., Sec. B, Vol. 62, Part 6, pp. 342-350, 1949.
2. May, K.R., "Spinning-top Homogeneous Aerosol Generator with Shockproof Mounting", J. Sci. Instr., Vol. 43, pp. 841-842, 1966
3. Davies, C.N., and P.K.P Cheah, "Spinning Generators of Homogeneous Aerosols", J. Aerosol Sci., Vol. 15, pp. 719-739, 1984
4. Davies, C.N., and P.K.P Cheah, "The Spinning-Top Aerosol Generator-Improving the Performance", J. Aerosol Sci., Vol. 15, pp. 741-751, 1984
5. Gussman, R.A., "A Further Development of the May Spinning Top Aerosol Generator", Am. Ind. Hyg. Assoc. J., Vol. 42, pp. 208-212, 1981.
6. Mitchell, J.P., and R.L. Stone, "Improvements to the May Spinning-Top Aerosol Generator", J. Phys. Sci. Instrum., Vol. 15, pp. 565-567, 1982
7. Gussman, R.A., A.M. Sacco and N.M. McMahon, "Design and Calibration of A High Volume Cascade Impactor", J. Air Poll. Cont. Assoc., Vol. 23, 1973

A STAG 2000 Caculator is available on the BGI Web page for the instrument, at <http://www.bgiusa.com> . Details of its use are explained, in section 4 .

When using the STAG 2000 there are two articles of paramount importance for its successful utilization: Cleanliness and attention to set up details. At the end of each session of use, time must be allocated for through clean up. This is especially important when solid aerosols are being produced by evaporative techniques, which leave copious residues. Every one of the several adjustments available on the instrument are of importance and cannot be ignored. However, leveling of the instrument carefully, outweighs all others.

Parts interchangeability with obsolete models: While the STAG 2000 bears an evolutionary resemblance to the STAG Mk I and Mk II, *there is no interchangeability of parts, with the following exceptions:*

1. The Disc is identical to the one used in the Mk I and Mk II as originally described by May⁽²⁾. However, it is manufactured from 303 S.S and will not activate the electronic tachometers used with the Mk II and some late Mk I units. (P/N A2214)
The Mk II utilized a hydraulic needle adjust mechanism as originally described by Mitchell⁽⁶⁾. This device has not been utilized on the STAG 2000. Its manual mechanism will completely replace the hydraulic components of the Mk II (P/N 10076).
4. The Mk II used a special elastomeric dampner to isolate the stator and rotor from external vibrations. This component of the STAG 2000, is interchangeable with the Mk II. (P/N A2222).

2.0 Principle of Operation.

When a disc is rotated, in a horizontal plane and a liquid is caused to flow onto the center of it, centrifugal force will cause the liquid to uniformly spread outward to the periphery of the disc. If the flow of liquid is not too great and the edge of the disc is sharp, ligaments will form. The ligaments will stretch, until the surface tension, of the liquid, is overcome, at which point it will snap off and a droplet will be formed. Approximately four satellite droplets will form from the tail of the ligament. Their diameter will be about 25% of the diameter of the primary droplet. Because the design of the current instrument follows the principles laid down by May⁽²⁾, the air flow which drives the disc is directed downwards. A cover is provided which channels this airflow within the body of the instrument. The force of this air flow is sufficient to entrain and remove the satellite droplets, whilst leaving the primary droplets trajectory, unaffected. This extraction effect is further enhanced by the provision of an air ejector provided on the body of the instrument. The body of the instrument is provided with a poppet valve which has the function of opening during start up and shutdown to provide stability to the disc.

Although the instrument can only aerosolize liquids, solid monodisperse particles may be formed by evaporation. Small solid particles, such as spores and bacteria, may be suspended in liquid and also dispersed as an aerosol.

3.0 Set Up

3.1 Open Model

The open model is the traditional instrument and the only type supplied commercially, heretofore. It produces its aerosol in a flat 360 degree spray in the plane of the disc. Figure 1, shows a schematic diagram of the components of the instrument set up for operation.

3.1.1 Important points to note:

The output of main disc air is connected to the *bottom* of the unit by pressing the provided tubing into the side connector. *Do not feed compressed air to the center connection.* The center connection is left open. Its function is to provide a vent for the poppet valve, under the disc.

The tube from satellite extraction air is connected to the side arm of the ejector nozzle, as shown in figure 1.

There is no preferred location for the pneumatic control module. There is no critical separation distance between the control module and the disk.

The body of the instrument must be absolutely level, to achieve operational stability. This is accomplished by removing the feed needle support spider and the disc. A level is laid across the top surface of the cover and the adjusting screws in the base are manipulated until the bubble is centered. The level is turned 90 degrees and the process repeated. *Be certain that level is then checked again in the original position.*

A source of clean dry compressed air, up to 40 psig and 4 cfm must be provided. For best results an infusion pump, capable of delivering up to 1 ml/min, of liquid should also be provided.

3.1.2 Arrangement and adjustment

The STAG 2000 may be set up in a fume hood, glove box or low velocity wind tunnel. If necessary, or desired, the base may be unscrewed from the body and a user supplied holding fixture. It is also worthwhile to set the base on a $\frac{1}{4}$ to $\frac{1}{2}$ inch thick soft silicone elastomer pad, to further dampen external vibrations. Once the unit is leveled, the disc may be simply set in the center of the stator and the spider rods installed in the three small holes provided in the support ring. *Lightly* tighten the set screws in the support ring. Next, install the needle guard assembly and use the three thumb screws provided in the spider assembly to exactly center the clear plastic cylinder over the disc. Now adjust the width of the gap between the top of the disc and the plastic cylinder to about 0.1 in. This height adjustment is accomplished by loosening the previously tightened screws in the support ring and sliding the spider rods up or down, until the correct gap is achieved. *This setting is only a starting point.* Adjustments up or down will have to be made dependent on disc speed and quality of aerosol produced. Once a final adjustment is achieved, the three stops (A2345) may be set, so that the same position may be easily regained.

The initial height of the needle above the stationary disk should be about 1/8 in. This is because the higher the air pressure to the disc, the higher the rotational speed and *the higher the disc will ride in the stator.* When in operation, the needle should be close enough to the disc so that the liquid is applied to the disc as a smooth flow *and not dropwise.* The exact setting will require experimentation. **Caution:** It is common practice to start with larger particle generation and progress to smaller (higher speed), Increasing air pressure will cause the disc to a higher position. If the needle setting is not raised, it will likely hit the needle tip at the new setting.

Liquid flow to the disc may be up to 1 ml/min. However, 0.5 ml/min is a better starting point and will yield earlier success. Liquid may be fed to the needle with a burette fitted with an adjustable stopcock. A far simpler technique is to utilize an infusion pump, set for the desired rate of flow.

An exploded diagram showing all the parts of the open form instrument, are shown in figure 3.

3.2 Enclosed Model.

The enclosed model is shown in figures 4-6 and is based on the publications of Mitchell⁽⁶⁾ and Gussman^(5,7). Inasmuch as larger aerosols travel a greater distance, upon generation, the enclosure should not be used for primary droplets greater than 10µm. Although this model permits the passage of purified air into the enclosure and aerosol laden air out, a greater amount of labor is engendered to set up and clean the enclosure. It should not be presumed that the enclosure will permit the generation of toxic or harmful substances without additional enclosure/ventilation. Because of the numerous ways in which the enclosure may be utilized, it is the task of the user to provide air flow through the enclosure.

3.2.1 Set up.

Loosen the three spider rods and withdraw the complete spider/needle support assembly. Slide the three rods, protruding from the bottom of the enclosure, into the three larger holes in the support ring. Remove the needle support assembly from the spider and place it in the large hole, provided for that purpose in the clear plastic cover of the enclosure. Centering and height adjustments of the feed needle are as described for the open model.

The purpose of the baffles in the enclosure are to provide a smooth laminar flow of air past the disc. This air flow is of increasing importance as smaller particles are generated. This topic has been dealt with, in some detail in references 2, 6 and 7. In this design a flow rate of 150 Lpm should not be exceeded.

4.0 Aerosol Generation

4.1 Direct

Direct generation means that a low vapor pressure bulk liquid is formed into a system of monodisperse droplets. The diameter of the droplets produced is governed by and calculated from the model⁽²⁾:

$$d_w = K/\omega(D\rho_l/T)^{0.5}$$

Where:

d_w = Diameter of liquid droplet – cm

ω = Angular velocity of the disc – radians/second

D = Disc diameter – cm ($D = 2.54$)

T = Surface tension of liquid – dynes/cm

ρ_l = liquid density – g/cm³

K = Constant of design ~ 4.5

The design constant of 4.5 is a starting point which, can be modified as experience is gained with a specific instrument, liquid and set of conditions.

Each individual STAG 2000 is supplied with calibration data for air pressure vs. disc speed. This information is furnished on a floppy disc as an Excel™ spread sheet. A calculator is provided so that pressure can be determined from speed required. The calculator has two tabs (at the bottom of the page). "Calibration" is the air pressure vs. speed of an individual, serial numbered unit. If the individual user wishes to re calibrate their instrument and is facile with Excel™, columns A/8-17 and C/8-17 are the ones for new data insertion. Columns B and D will automatically be updated. The graphs were plotted with Jandel Tablecurve™ and are fixed. They can be deleted and replotted in Excel.

After deciding the primary droplet diameter, which is to be produced, enter this number in B/37. Enter the liquid properties in C/37 and D/37. The required rotational speed will be returned in A/37. Enter a pressure in A/27 or A/30 and change values until a number close to A/37 is found. This will be the pressure required to produce the nominal diameter desired.

4.2 Indirect

Indirect generation means that a low vapor pressure bulk liquid is formed into a system of monodisperse droplets (solvent), into which is dissolved the desired aerosol material (solute). The diameter of the droplets produced is still governed by and calculated from the model. Insert the desired dry particle diameter in B/44 and the balance of the solvent properties in C, D and E/44. The required primary droplet diameter will be returned in A/44. This figure may now be used to calculate the required rotational speed (A/37) and thus the air pressure required.

Bear in mind that these figures are all, nominal and are intended to minimize start up effort. Once sufficient data has been gathered, an investigator may solve the model equation for a more exact constant.

4.4 Concentration

A calculator is also provided for number and mass concentration. This is the page tab marked "Concentration". Enter the diameter of the primary droplet in A/7. Enter the liquid feed rate in D/7 and the number of droplets produced per minute will be returned in D/9. This will also be the number of dry particles per minute.

Similarly, if the diameter of the droplet (or dry particle) is entered in A/21, the total air flow in B/21 and the density of the aerosol material in C/21 the mass concentration produced will be returned in B/24.

4.5 Starting

Set an initial extraction air pressure of 3 psig. Set an initial disc pressure of 10 psig. Adjust the disc pressure to the desired value. If the disc exhibits any instability, reduce the pressure, allow it to stabilize and increase by small stages.

5.0 Cleaning

Frequency of cleaning is dependent upon the materials being generated. When utilized with low vapor pressure materials, very little needs to be done on a daily basis, unless instability of production begins to appear. However even with materials of this type at least weekly cleaning is required as many low vapor pressure materials plasticsize upon prolonged exposure to the air. All other material usage, requires some degree of cleaning on a daily basis. If an ultrasonic cleaner is not available for cleaning, one should be obtained.

Remove the needle holder and needle and clean. Loosen the three support spider binding screws (10037) and lift the entire spider, intact and clean. Lift out the disc and clean. Be especially careful not to drop this component lest the sharp, droplet formation edge becomes bruised. Unless encrustations of material are observed in the stator cup (B2213), this is sufficient cleaning.

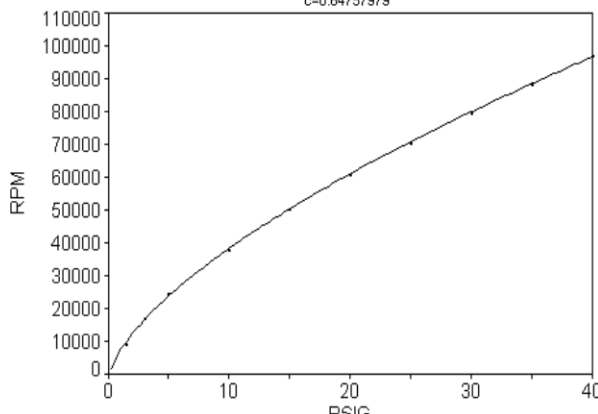
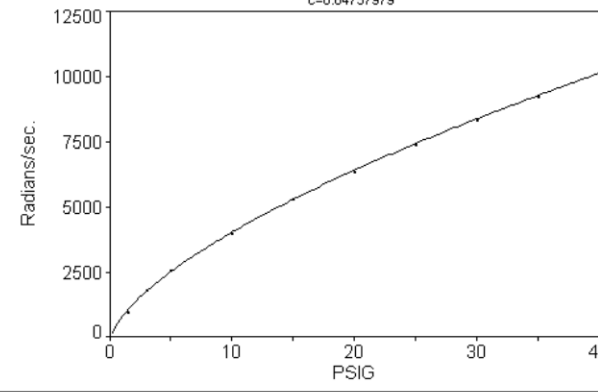
The next phase of cleaning requires detail disassembly. After the first time this operation is performed and confidence is gained, this operation can be performed in a very short time.

Remove the elbow (10042) from the vent tube on the underside of the unit. This is accomplished by pressing the collar surrounding the tube, inwards, towards the elbow. A pulling tension should be maintained on the elbow, during this operation. All connectors used on the body of the instrument are of this collet type. They are very convenient for rapid assembly/disassembly. Next, loosen the two set screws (10067), which secure the vent and feed tube connectors (10045), to the base of the instrument. Grasp the Stator base (B2215) with one hand and the base of the instrument with the other and withdraw the stator assembly from the body of the unit. The stator assembly can be ultrasonically cleaned and all passages blown clear with compressed air or disassembled in detail, per Figure 3. *(Caution. When clearing passages of liquid, by the use of compressed air, an aerosol may be generated which is harmful. Use good laboratory safety precautions)*

Re assembly is in the reverse order of assembly. Tubing is reinstalled in the various collet connectors by pushing in until the internal stop is reached. However, it is *crucially important* that a slight compression be placed on the elastomeric damping section. This is accomplished, by pressing down, lightly, on the stator assembly while tightening the set screw (10067) which secures the *center* collet connector (10045) to the base.

6.0 Revision History

Initial composition	February 2002
Added Warning on page 1	November 2005

	A	B	C	D	E	F	G	H	I	J	K						
1	Stag - 2000 Calibration			Serial No.	A001	Pressure vs. RPM Rank 1 Eqn 8010 [Power] $y=a+bx^c$ $r^2=0.99975824$ DF Adj $r^2=0.99963736$ FitStdErr=542.1371 Fstat=14473.775 $a=-2053.039$ $b=9023.9731$ $c=0.64757979$ 											
2																	
3	Poppet valve: SS			Date:	15-Nov-01												
4	Extraction air pressure 3psig			Operator:	rg												
5																	
6	Pressure	Pressure	Revolutions/	Radians/													
7	PSIG	kPa	Minute	Second													
8	40	275.6	96673	10124													
9	35	241.2	88125	9228													
10	30	206.7	79448	8320													
11	25	172.3	70294	7361													
12	20	137.8	60469	6332													
13	15	103.4	50291	5266													
14	10	68.9	37541	3931													
15	5	34.5	24276	2542													
16	3	20.7	16933	1773													
17	1.5	10.3	8917	934													
18																	
19	All information to be user entered is red. All answers are blue																
20																	
21	Note: The rotational speed caculator is specific to an																
22	individual instruments calibration																
23																	
24	Rotational Speed Calculator																
25	PSIG	R/sec				Pressure vs. Radians/sec. Rank 1 Eqn 8010 [Power] $y=a+bx^c$ $r^2=0.99975824$ DF Adj $r^2=0.99963736$ FitStdErr=56.772464 Fstat=14473.775 $a=-214.99374$ $b=944.98825$ $c=0.64757979$ 											
26																	
27	20	6361															
28	kPa	R/sec															
29																	
30	100	5128															
31																	
32	Primary droplet diameter																
33																	
34	Rotation	Primary	Liq. Surface	Liquid	Design												
35	Radians/	Droplet dia.	Tension	Density	Constant												
36	Second	μm	dynes/cm	g/cm3	K												
37	715	58.4	22.7	1	4.5												
38																	
39	Dilution Concentration Calculator																
40																	
41	Wet	Dry	Solute	Solvent	Conc.												
42	Droplet dia.	Droplet dia.	Density	Density	%												
43	μm	μm	g/cm3	g/cm3	by weight												
44	58.4	10	1.26	0.789	0.008												
45																	
46																	
47																	
48																	
49																	
50																	
51																	
52																	
53																	
54																	
55																	
56																	
57																	
58																	

	A	B	C	D
1	STAG 2000 Number Concentration Calculator			
2				
3	Note: The computations are based upon satellites whose diameter is 1/4 that of the primary droplet.			
4	It is assumed that there are four satellites/droplet.			
5				
6	Dia. μm	Volume	cm^3	Liquid feed rate-cm^3/min
7	10	Primary Droplet Volume	7.8540E-10	1
8	2.5	Satellite Volume	1.2272E-11	No. Conc. of primary droplets/min
9		4X Satellite Volume	4.9087E-11	1.20E+09
10				
11	Enter the diameter of a primary droplet and the number concentration of primary droplets/min is returned.			
12				
13	Enter the liquid feed rate and the number concentration of primary droplets/min is returned.			
14				
15	STAG 2000 Mass Concentration Calculator			
16				
17	Before using this section, complete the one above. <i>This section is dependent</i>			
18	Enter the total air flow and the density (g/cc) of the aerosol material and the mass concentration is returned			
19				
20	Dia. μm	Total Air Flow - Lpm	Density of Aerosol material	No. Conc. of primary droplets/min
21	10	10	1	1.20E+09
22				
23		Mass Concentration mg/L		
24		94.1		
25				
26				
27				

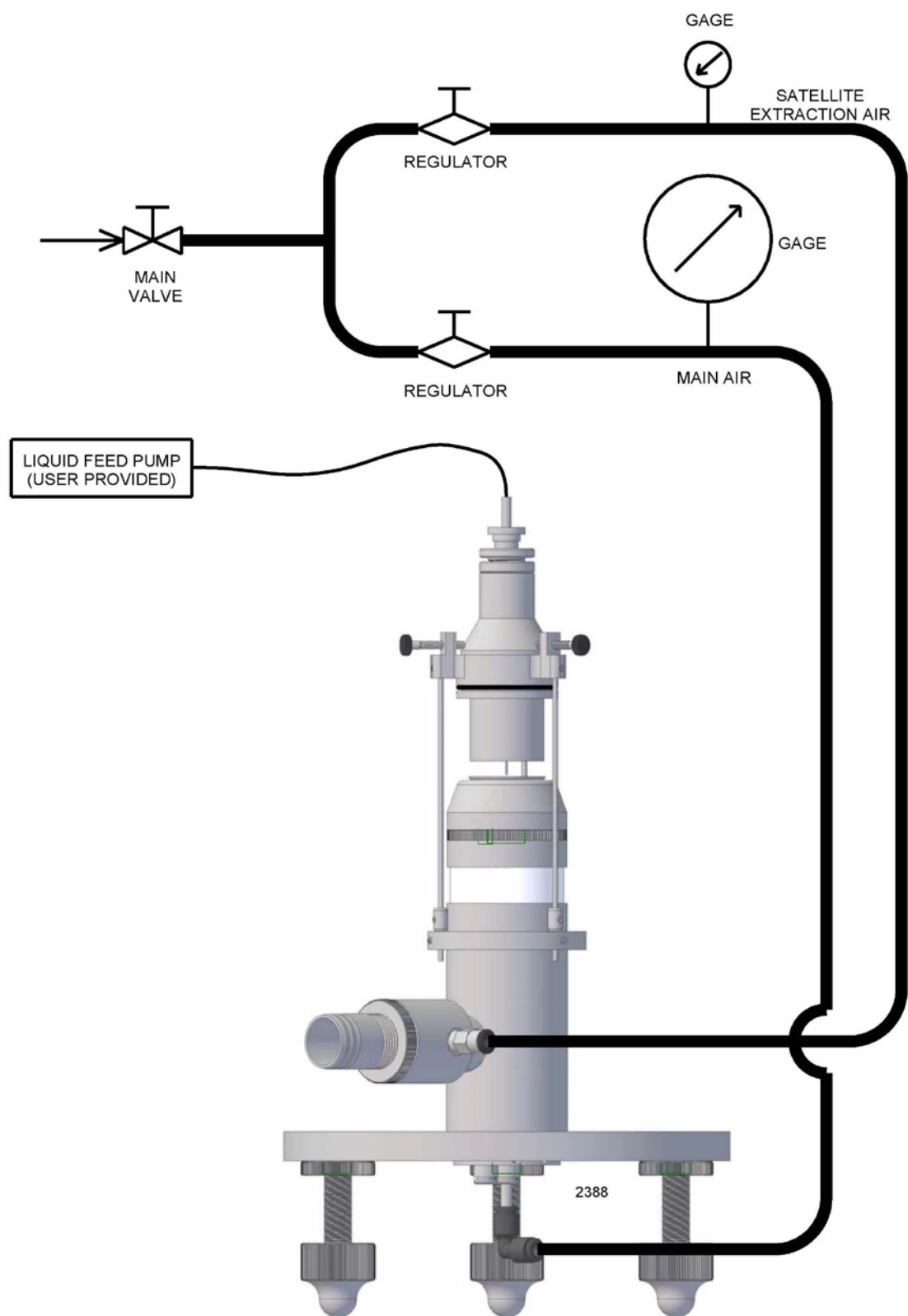


Figure 1: Schematic diagram of STAG 2000 General Setup

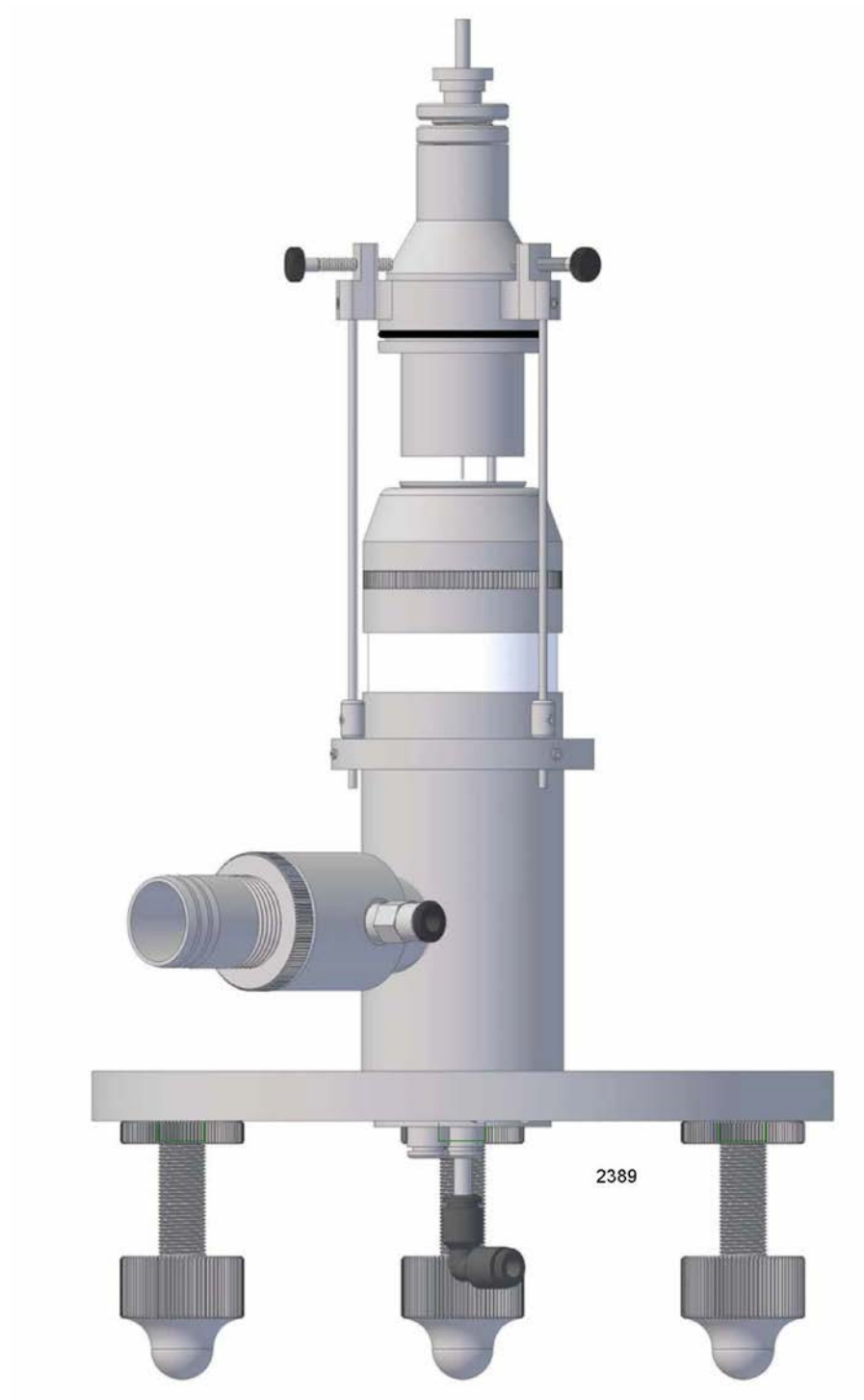


Figure 2: Diagram of STAG 2000

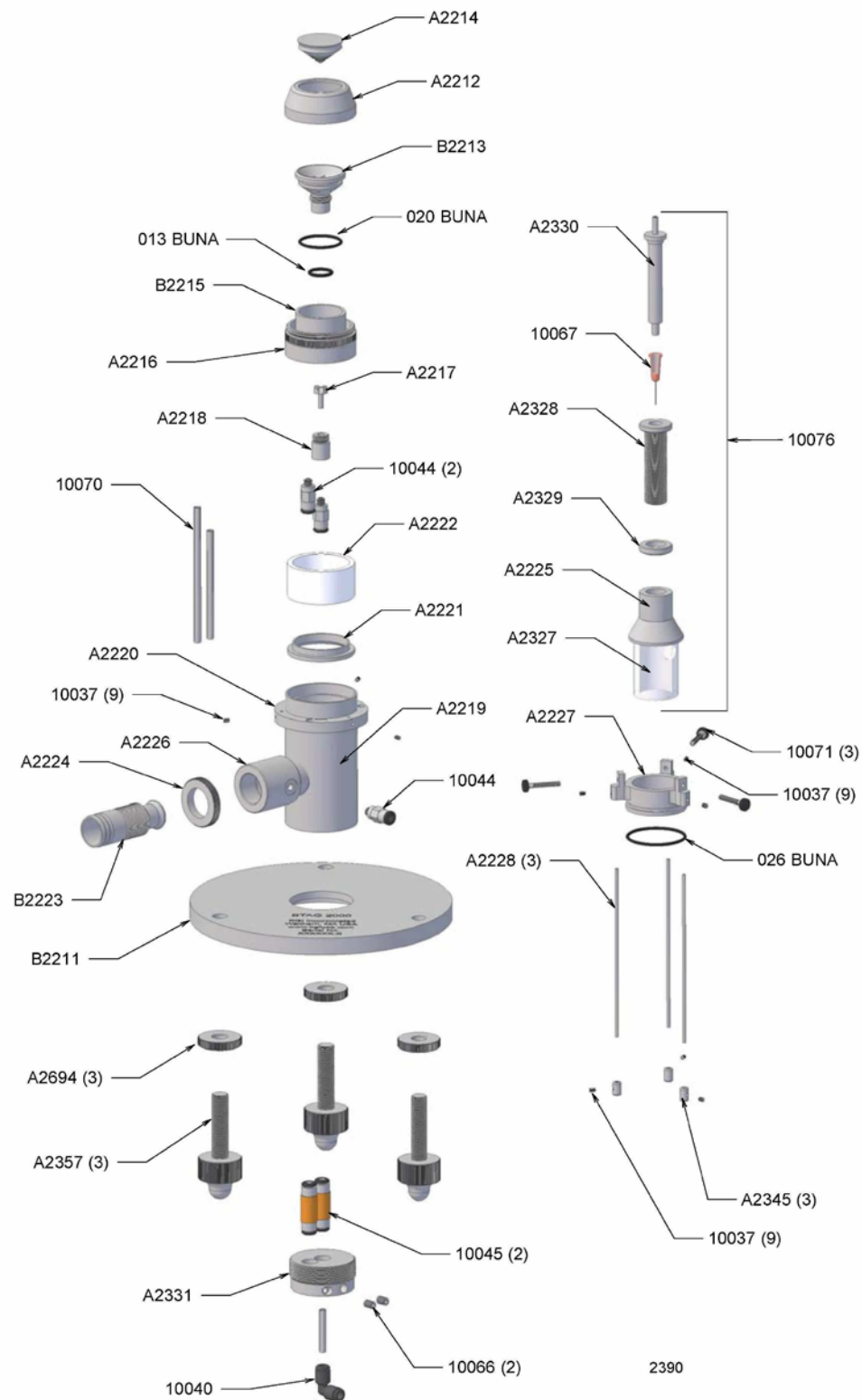


Figure 3: Exploded View of STAG 2000

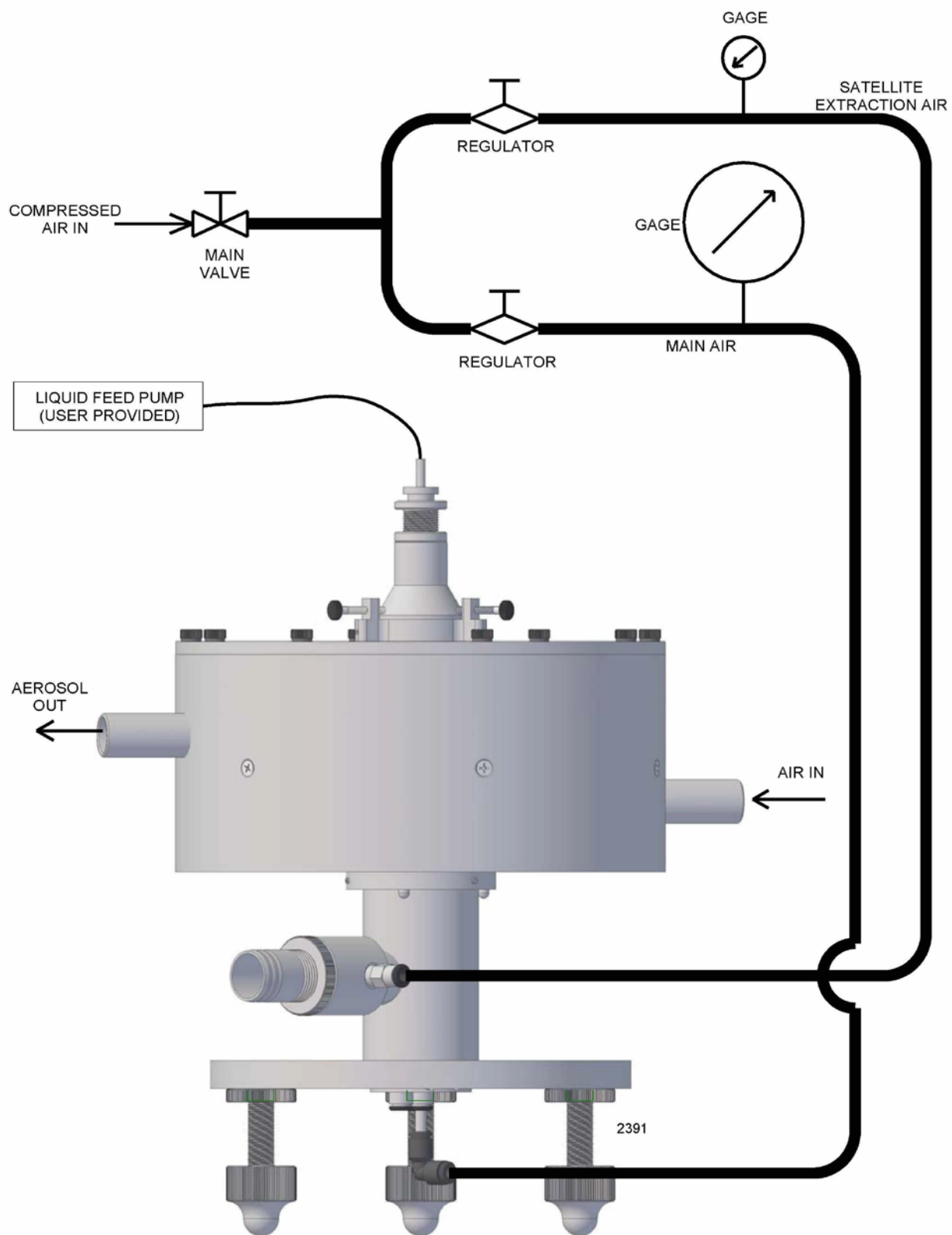


Figure 4: Schematic Diagram of STAG 2000 General setup. Enclosed Model

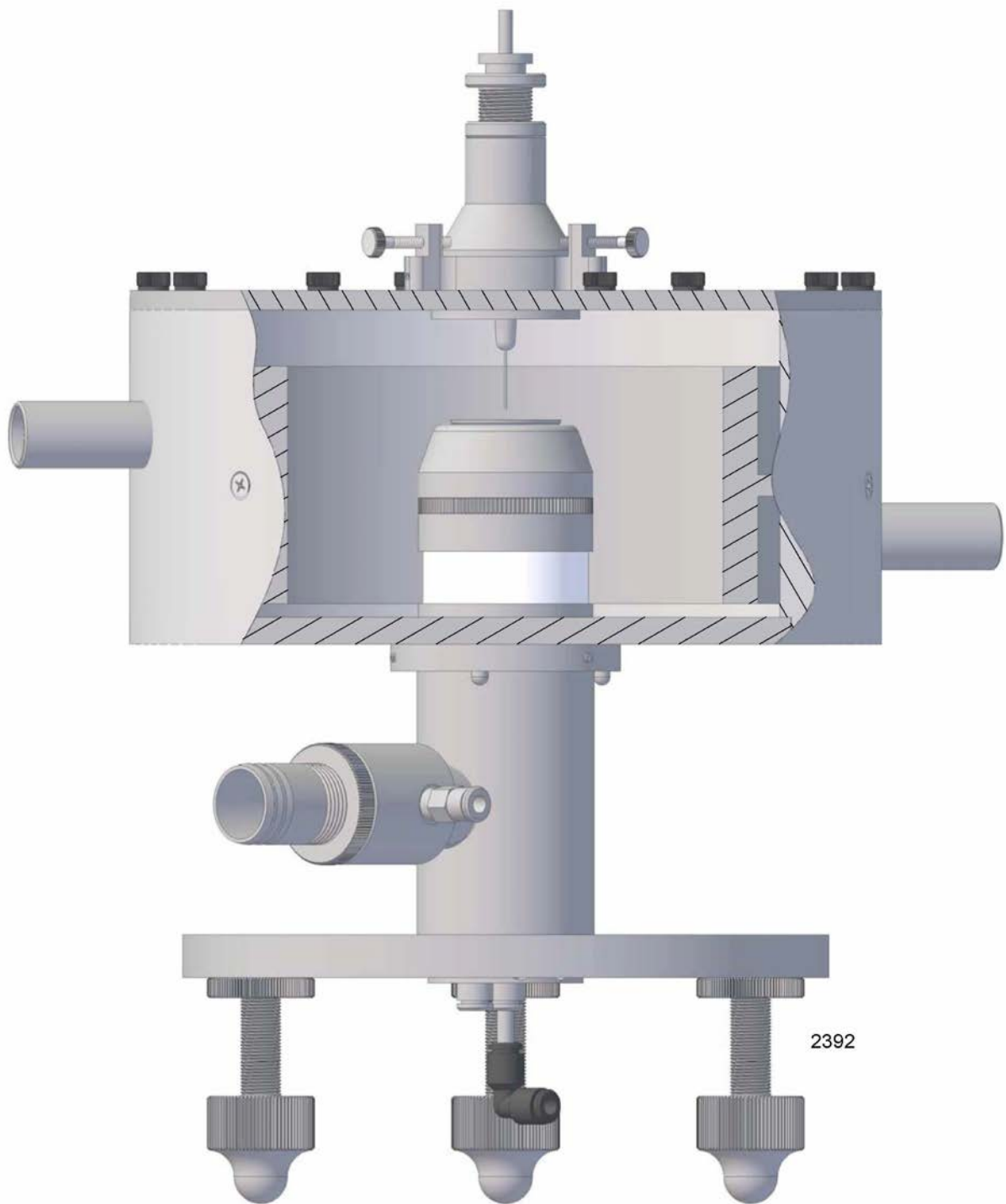


Figure 5: Diagram of Enclosed STAG 2000

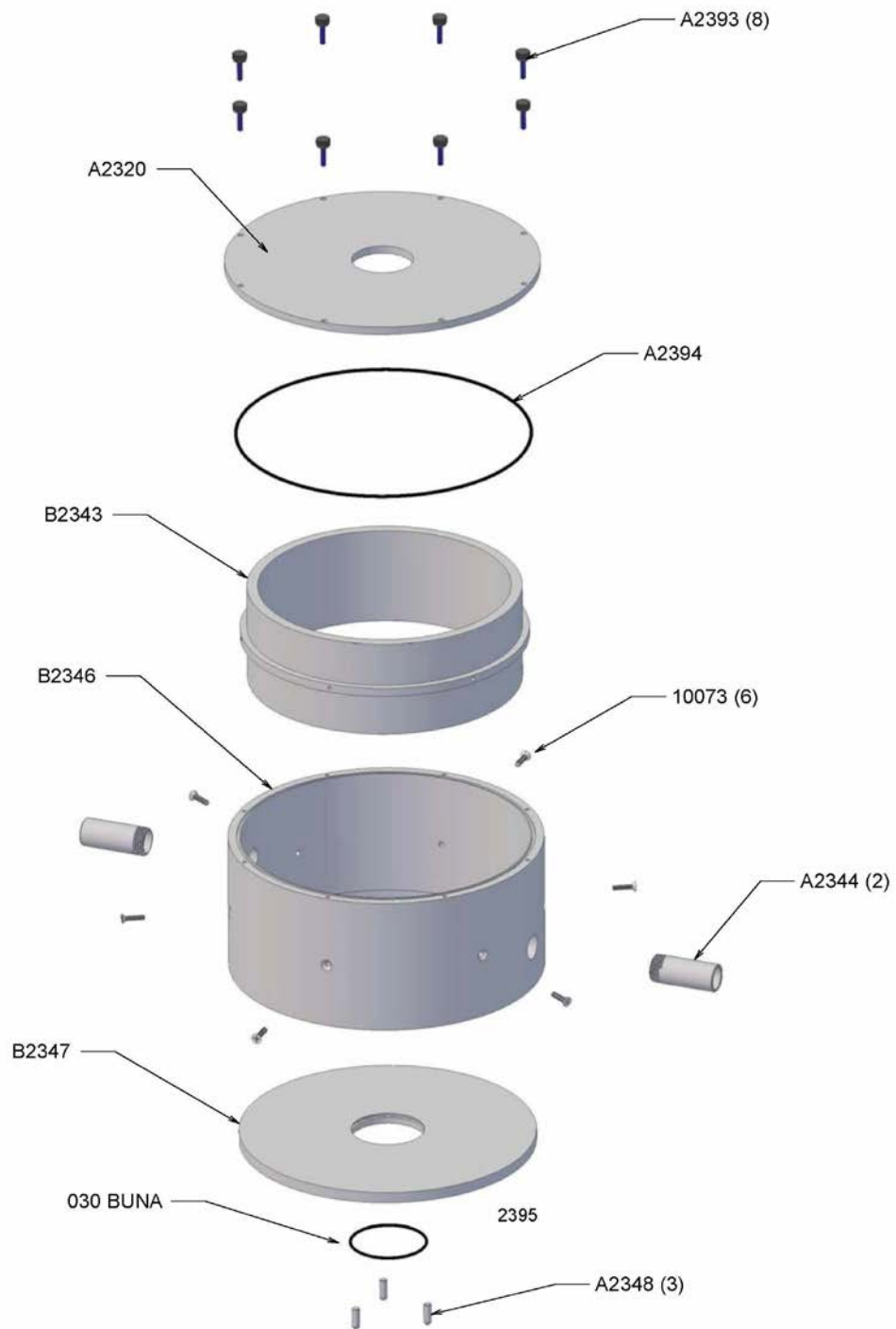
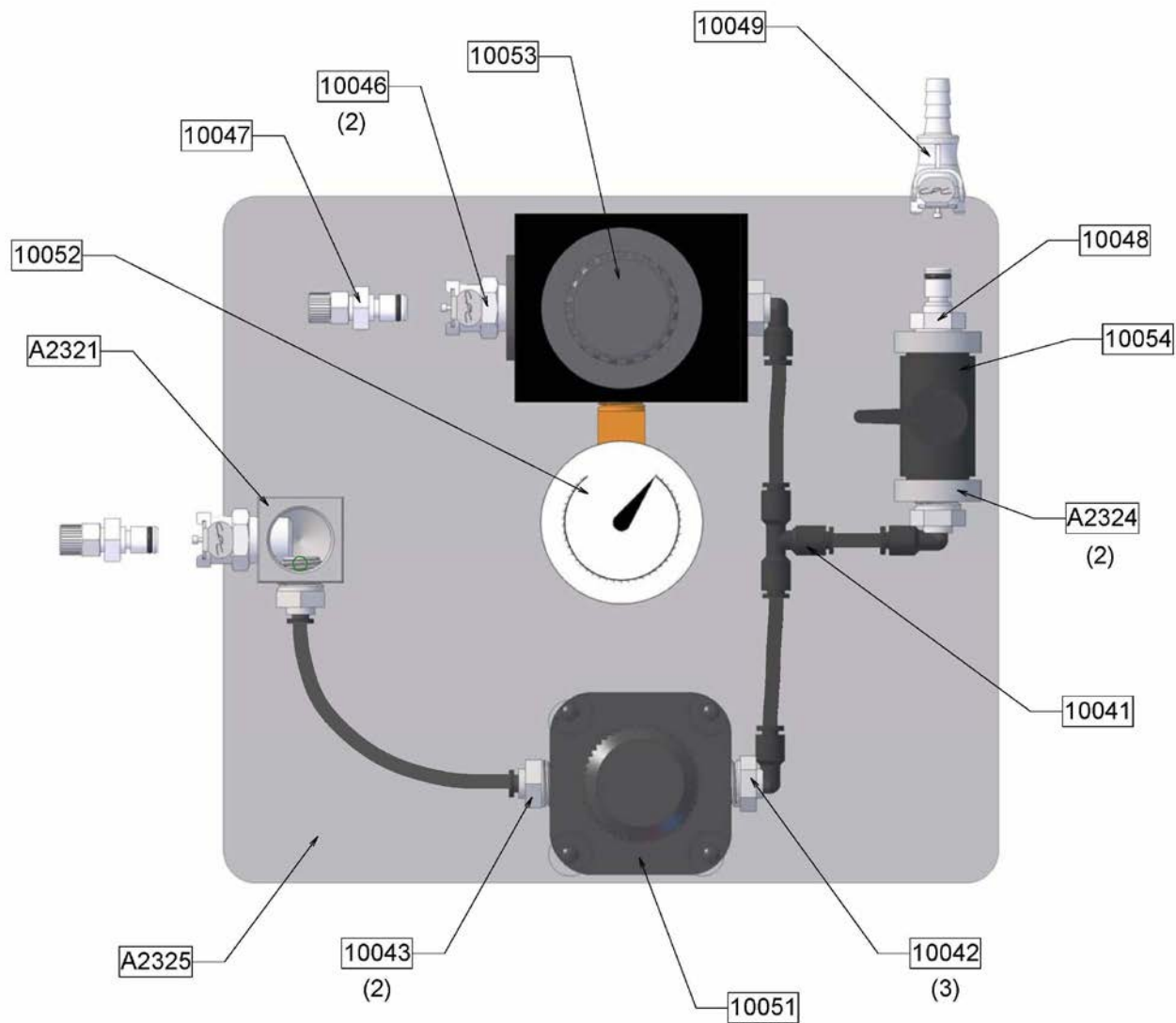


Figure 6: Exploded View of STAG 2000 Enclosure



Notes:
 1) Main Gage Not Shown - P/N 10050
 2) Complete Control Module - P/N 10077

2399

Figure 7: Location of Control Module Components

PART #	FIG.	QTY.	DESCRIPTION
10037	3	9	#2-56 x 1/8" Set Screw
10040	3	2	Push-to-Connect Elbow
10041	7	1	Push-to-Connect Tee
10042	7	3	Push-to-Connect Male Elbow
10043	7	2	Push-to-Connect Male Connector
10044	3	3	Push-to-Connect Fitting
10045	3	2	Push-to-Connect Union
10046	7	2	Colder 1/4" Fitting W/Shutoff
10047	7	2	Colder In-Line Fitting
10048	7	1	1/4" NPT Coupling Insert
10049	7	1	1/4" Colder Hose Barb Fitting
10050	7	1	Main Gage
10051	7	1	Main Regulator
10052	7	1	Auxiliary Gage
10053	7	1	Auxiliary Regulator
10054	7	1	Ball Valve
10055	-	6	#8-32 x 3/8" Socket cap Screw SS
10056	-	6	#8-32 x 1 1/4" Socket cap Screw SS
10066	3	2	#8-32 x 1/4" Set Screw SS
10067	3	1	Needle
10070	3	1	5/32 x .106 Nylon Tubing (14')
10073	6	6	#4-40 x 1/2" Flat Head Socket Screw SS
10076	3	1	Complete Manual Needle Adjuster
10077	7	1	Complete Control Module
013 Buna	3	1	O-Ring
020 Buna	3	1	O-Ring
026 Buna	3	1	O-Ring
030 Buna	6	1	O-Ring
A2212	3	1	Spinning Top Cover
A2214	3	1	Rotor
A2216	3	1	Balance Ring (also order P/N B2215)
A2217	3	1	Poppet Valve
A2218	3	1	Poppet Housing
A2219	3	1	Body Tube (also order P/N's A2220 & A2226)
A2220	3	1	Tri-Stand Ring (also order P/N's A2219 & A2226)
A2221	3	1	Ring
A2222	3	1	Dampner
A2224	3	1	Outlet Nut
A2225	3	1	Adjuster Body (also order P/N A2327)
A2226	3	1	Exhaust Tube (also order P/N's A2219 & A2220)
A2227	3	1	Spider Mount

PART #	FIG.	QTY.	DESCRIPTION
A2228	3	3	Spider Leg
A2320	6	1	Shroud Top
A2321	7	1	Gage Standoff
A2322	-	2	Auxiliary Regulator Standoff
A2323	-	4	Main Regulator Standoff
A2324	7	2	Valve Standoff
A2325	7	1	Base Plate
A2327	3	1	Adjustor Tube (also order P/N A2225)
A2328	3	1	Needle Adjustor
A2329	3	1	Nut
A2330	3	1	Needle Holder
A2331	3	1	Tubing Mount
A2344	6	2	Inlet/Outlet Tube
A2345	3	3	Stop
A2348	6	3	Pin, Bottom Plate
A2357	3	3	Leg
A2393	6	8	Screw
A2394	6	1	O-Ring
A2694	3	3	Leg Nut
B2211	3	1	Base
B2213	3	1	Stator
B2215	3	1	Connector (also order P/N A2216)
B2223	3	1	Exhaust Tube
B2343	6	1	Collection Chamber Inner Ring
B2346	6	1	Collection Chamber Outer Ring
B2347	6	1	Collection Chamber Bottom Plate